Braden Wei

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Education

McGill University **Montreal, QC**

B.Sc Computer Science *September 2023 — May 2027*

- **Coursework**: Computer Networks, Operating Systems, Algorithms & Data Structures, Probability & Statistics
- **Hack McGill:** Logistics & Sponsorship Coordinator coordinated $35,000 sponsorships for 600+ participants

Experience

Crusoe **San Francisco, CA**

Software Engineering Intern - Crusoe Managed Kubernetes *June 2026 - Sep 2026*

- Owned the design and launch of node pool autoscaling as a first-class customer-facing API feature across **7 repositories**, spanning protobuf schemas, Go services, Kubernetes management, client libraries, CLI, Terraform, and the cluster autoscaler.
- Moved autoscaling bounds from internal Kubernetes deployment flags into the node pool API, giving customers direct min/max node controls, enabling scale-to-zero, and preventing unexpected unbounded scaling.
- Cut node pool failover from **15 minutes to 41 seconds (22x)** by designing a state-based failover path, surfacing scale-up failures to cluster autoscaler, triggering exponential backoff and automatic fallback to healthy pools.
- Built VM and node-pool quota enforcement for a new provisioning flow in **Go** and **Temporal**, acquiring capacity permits before allocation and surfacing quota failures as actionable customer health issues.
- Authored the autoscaler compatibility design for a major node-pool architecture migration, identifying deletion and rollout failure modes that could affect existing customers and driving adoption of a **targeted node-removal API**.

Zebra Technologies **Montreal, QC**

Software Engineering Intern - Aurora Design Assistant *January 2026 - April 2026*

- Built **C#** and **Selenium** end-to-end test automation for Aurora Design Assistant, Zebra's machine vision IDE.
- Benchmarked acquisition across 5 camera models over GigE and USB3 Vision, tracking CPU/RAM usage to separate software bottlenecks from hardware limits.

Valsoft Corporation **Toronto, ON**

AI Engineer Intern (Part-Time) *Sep 2025 - Dec 2025*

- Upgraded production RAG backend with **file-search tool calling** and token optimization, reducing **hallucination rate by 35%** and improving **response accuracy by 20%**
- Built a real-time management dashboard by integrating 40+ Electronic Healthcare Record modules with a Java REST API and Supabase webhooks, enabling usage tracking, adoption monitoring and churn prediction

AI Engineer Intern *June 2025 - August 2025*

- Built an AI assistant for healthcare professionals using hybrid RAG, reducing manual support and form completion errors by **60%**.
- Developed a HIPAA-compliant backend integrating Java EHR services with FastAPI inference, achieving **sub-3 second** response latency through caching and token optimization.
- Instrumented the pipeline with **Langfuse** for tracing, prompt versioning, and eval datasets, tracking retrieval quality and hallucination rates across 10+ healthcare facilities.

Projects

Roomiez | *React Native, Expo, Express.js, Firebase*

- Built a cross-platform roommate-matching app with React Native, Express.js, and Firebase, implementing secure authentication, REST APIs, and real-time profile and chat synchronization.

Technical Skills

Languages: Go, Python, Java, TypeScript/JavaScript, SQL, C++
Cloud & DevOps: Kubernetes, Terraform, GCP, Docker, Helm, GitHub Actions, CI/CD
Backend: PostgreSQL, Redis, gRPC, Prometheus, Grafana
Frameworks & Tools: React, Next.js, Node.js, FastAPI, Temporal, Git, Linux